

ADDRESSING A BLIND SPOT IN A TOPOLOGICAL DATA ANALYSIS–BASED DATA POISONING DETECTOR

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Motivation

Data poisoning can insert attack-derived traffic with benign labels, shifting a learned decision boundary before deployment. Monkam et al. [1] propose an upstream TDA–OPTICS filter to identify poisoned packets before model fitting.

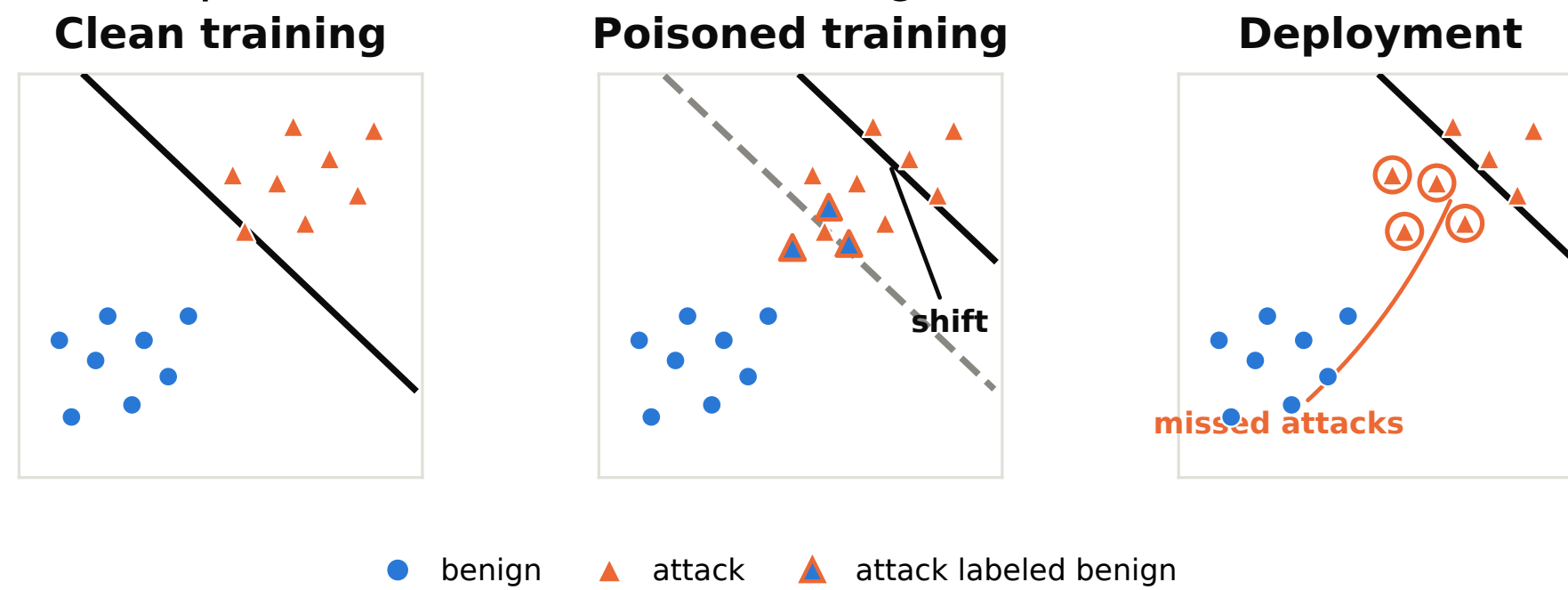


Figure 1. Data poisoning can cause mislabeling, which can shift a learned boundary and degrade model performance.

Persistent Homology

Persistent homology tracks connected components (H_0) and loops (H_1) across filtration scales; long intervals encode durable structure.

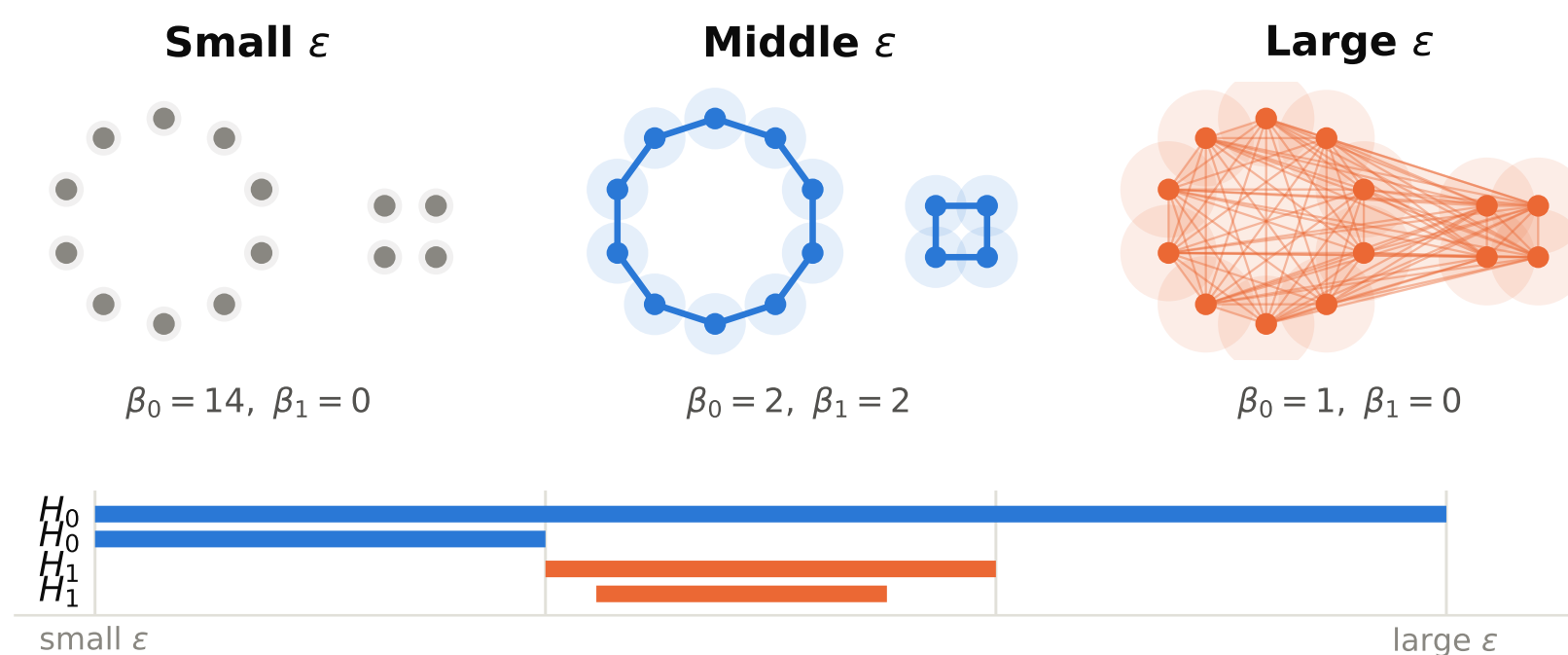


Figure 2. Point-cloud neighborhoods and the corresponding H_0/H_1 barcode as the scale ϵ increases.

Reconstruction and Controlled Test

Data: UNSW-NB15 Payload-Byte [2].

1,500 bytes $\rightarrow$ 30×50 raster $\rightarrow$ $\text{Bin}_t \rightarrow$ 2 height + 3 radial filtrations $\rightarrow$ cubical persistence $(H_0, H_1) \rightarrow \mathbb{R}^{60} \rightarrow$ OPTICS

- Four probe families $\times$ five fixed seeds; each cell has 5,000 clean + 500 poison packets. Values = means $\pm$ population SD.
- 4 permutation families: Transpositions (Swap 60 pairs of random bytes), Block Reversal (Mirror 120-byte block), Block Swap (Swap two 60-byte blocks), Cyclic Shift (Rotate every row by 1 index).
- Every probe changes the raw vector and preserves its byte multiset.
- Ground-truth labels are used for scoring only.

TDA Pipeline

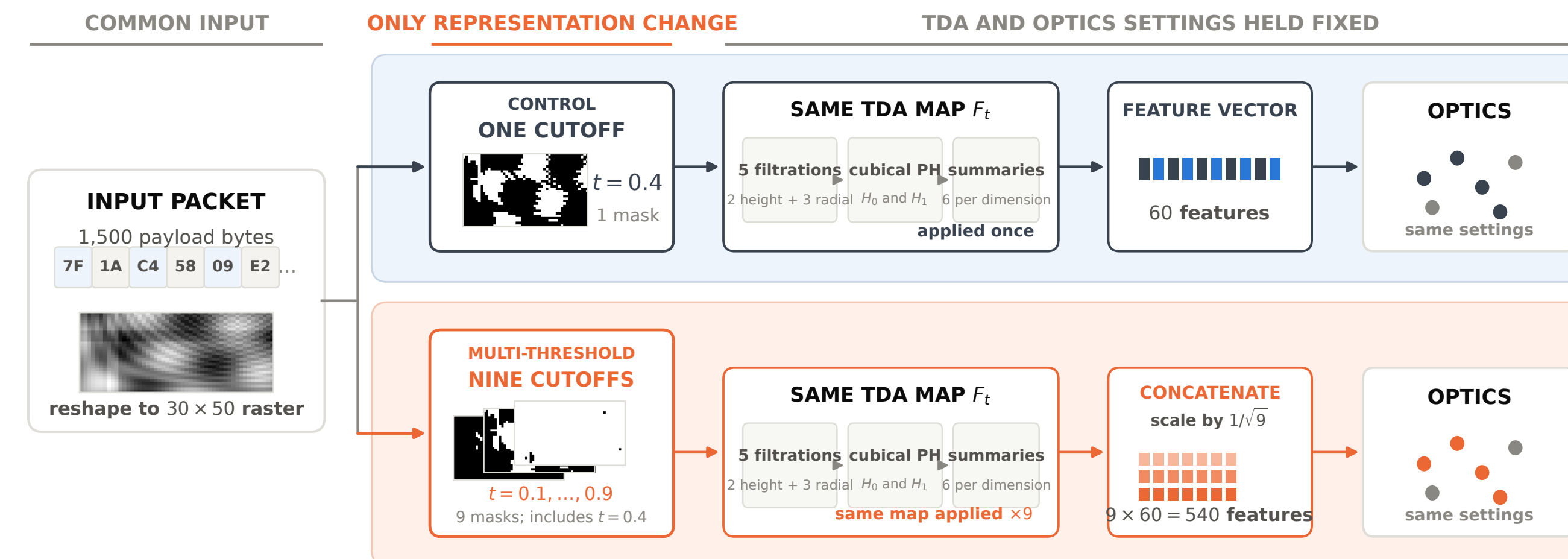


Figure 3. Only the threshold representation changes. Each cutoff uses the same five-filtration, 60-feature TDA map; the nine resulting blocks are concatenated and scaled before the unchanged OPTICS configuration.

Results

92–96% Fewer Poisoned-Clean Vector Collisions using a multiple-threshold approach.

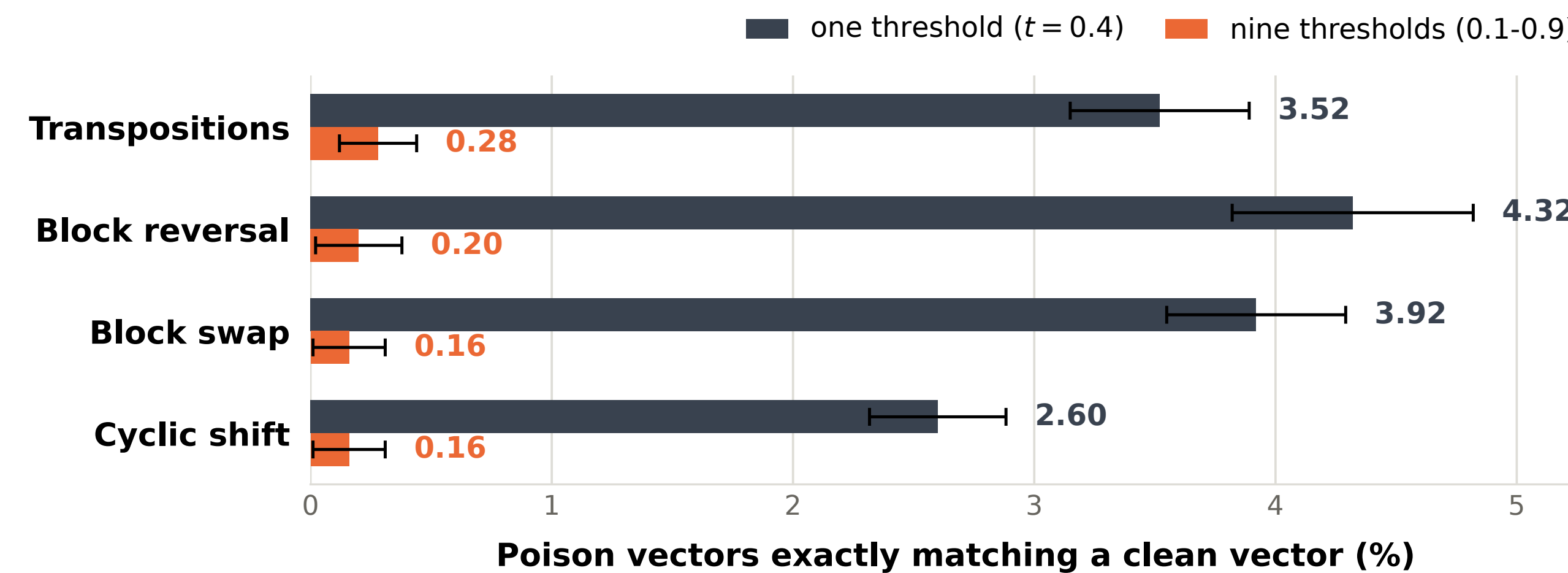


Figure 4. Five-seed Phase Q means for 5,000 clean + 500 raw-changing poison packets per cell. Exact poison–clean matches fall by 92–96%.



Figure 5. Separate legacy-R60 audit: 97.5% of repeated-vector collision mass first merges at raw payload or binarization. This is not the Phase Q population.

References and Contact

- [1] G. F. Monkam, M. J. De Lucia, and N. D. Bastian, *Computers & Security* 144, 103929, 2024.
[2] N. Moustafa and J. Slay, “UNSW-NB15,” *MilCIS*, 2015.
[3] M. Ankerst et al., “OPTICS,” *SIGMOD*, pp. 49–60, 1999.



Why One Cutoff Loses Information

Let $z = x/M$, and let S_n permute byte positions. Permutations preserve the fitted maximum M , so thresholding $\text{Bin}_t(z)_i = 1\{z_i > t\}$ is equivariant. At $t = 0.4$, let

$$A^- = \{i : z_i \leq 0.4\}, \quad A^+ = \{i : z_i > 0.4\}.$$

Then

$$G_z := \text{Stab}(z) \leq G_{0.4} := \text{Stab}(\text{Bin}_{0.4}(z)) = S(A^-) \times S(A^+).$$

The meaningful blind spot is $G_{0.4} \setminus G_z$: raw-changing permutations erased by the cutoff. It is nonempty precisely when one threshold class contains two distinct byte values. For $T = \{0.1, \dots, 0.9\}$, define

$$Q_T(z)_i = (\text{Bin}_t(z)_i)_{t \in T}.$$

If $B_0, \dots, B_9$ are the induced bands, then

$$G_z \leq G_T := \text{Stab}(Q_T(z)) = \prod_{k=0}^9 S(B_k) \leq G_{0.4}.$$

The final containment is strict when an occupied 0.4 class spans multiple bands. Within-band permutations and later noninjective maps can still create collisions. Thus the stack reduces representation invariance; it does not guarantee detection.

Controlled representation change

Control: $F_{0.4}(x) \in \mathbb{R}^{60}$,

Repair: $F_T(x) = \frac{1}{\sqrt{9}}(F_{0.1}(x), \dots, F_{0.9}(x)) \in \mathbb{R}^{540}$.

Each F_t uses the same 60-feature pipeline. The same packet pairs enter both arms; $1/\sqrt{9}$ matches a single-block Euclidean distance only when all nine blockwise differences are identical.

Conclusion and Scope

- Nine thresholds reduce exact poison–clean collisions by 92–96%. Multi-threshold technique is empirically shown to cover a blind spot in the original study.
- However, Matched-cost data poisoning capture does **not** materially improve $\rightarrow$ Repair mainly fragments OPTICS output and sends poisoned data to noise, leaves unclustered.
- Future work should improve attack separation relative to ordinary clean variation before retuning or replacing OPTICS.